

Encrypted certificate generation for NHCX participants (*X.509 RSA Certificate*)

Step 1 - Generate a 2048-bit RSA Private Key

```
openssl genpkey -algorithm RSA -out private.key -pkeyopt rsa_keygen_bits:2048
```

This will output a private.key (RSA Private Key).

Step 2 - Create a certificate signing request

```
openssl req -new -key private.key -out request.csr
```

This will prompt for country, state, organisation etc and output a request.csr.

Step 3 - Generate a Self-Signed X.509 Certificate

```
openssl x509 -req -in request.csr -signkey private.key -out certificate.crt  
-days 365
```

This will output X.509 format a self signed RSA certificate certificate.crt

Sample X.509 format self signed RSA certificate

-----BEGIN CERTIFICATE-----

MIID0zCCArugAwIBAgIUaxmygMvtb6JvQyB19mvjr2cd7fowDQYJKoZIhvcNAQEL
BQAweTELMakGA1UEBhMCSU4xEjAQBgNVBAGMCUpoYXJraGFuZDEQMA4GA1UEBwwH
RGhhbmJhZDELMAkGA1UECgwCTkExCzAJBgNVBAsMAk5BMQswCQYDVQQDDAJQTEd
MBsGCSqGSIb3DQEJARYOdGVzdEB1bWFPbC5jb20wHhcNMjUwNzEzMTkyMDA4WhcN
MjYwNzEzMTkyMDA4WjB5MQswCQYDVQQGEWJTTjESMBAGA1UECAwJSmhhcmtOYW5k
MRAwDgYDVQQHDAdEaGFuYmFkMQswCQYDVQQKDAJOQTELMakGA1UECwwCTkExCzAJ
BgNVBAMMAk5BMR0wGwYJKoZIhvcNAQkBFg50ZXN0QGvtYWlsLmNvbTCCASIwDQYJ
KoZIhvcNAQEBBQADggEPADCCAQoCggEBAK0dFMDNq1AmLICcy2CKcA27Da+dEUj2
VhTv5eSRfS5vq+2IJ30hRyMvGHL3ReqdH0SPiOLzbPH9pe2AdUBdWMv+uoaGumZo
c3+G1+3f1Mu/i2GYJpLSG2p781egxXbqSTaHHlqT46KIz3b+E0+dWtkC6T6PWIdQ
S4eNTZzVW+xErWcVy/+MhnNkGzpr7H1Tuj8lyRm9ZofvVioVQEo0lcbdtHRRHw1
fgUu6WubGypIcvZi2LjbJ6D+bdwtu/o661zmndyo+T6FSIa+AHEgUSWILx0+QJB
iY0b59UNBKvQe7Sr5E+8bNE2nQ8n9GAKmDFQhUoTfBiA25hFka8Q7V0CAwEAAANT
MFEwHQYDVR00BBYEFKMAjFPnbVSbRW5UuN01AZnsypgMB8GA1UdIwQYMBaAFLKM
AjFPnbVSbRW5UuN01AZnsypgMA8GA1UdEwEB/wQFMAMBAf8wDQYJKoZIhvcNAQEL
BQADggEBAAdzgVXWU1z1SQqv0Fi8KWs9dtT+mMA0nwz5Y68Z95kbHqB3UC9VLGsu
wJIXPMQ/ViojN7IjLOK3uTr6BjbNdI0fVIzJHFPHYuWmQGIa7keHh7egBbAkp1EG
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3T289PUZexmJE1zqksy5gYlyiujdxaJGLPkQ/FV9vynZyc9S7jBBmHmttDw8BaVX
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5K09n3b+37/gDJA/EzJLyuNLSypykPU=

-----END CERTIFICATE-----

Step 4 - Base64 Encoding of generated RSA certificate

Using any base64 encoder, for example [Base64 Encoder](#), generates a base64 encoded encrypted certificate.

```
LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSUQwekNDQXJ1Z0F3SUJBZ0lVYXhteWdNdnRiNkp2
UX1CMT1tdmpyMmNkN2Zvd0RRWUpLb1pJaHZjTkFRRUwKQ1FBd2VURUxNQWtHQTFVRUJoTUNTvTR4RWpB
UUJnTlZCQWdNQ1Vwb1lYSnJhR0ZlWkRFU01BNEdBMVVFQnd3SAPSR2hoYm1KaFpERUxNQWtHQTFVRUNn
d0NUa0V4Q3pBSk1JnTlZCQXNNQWw1Qk1Rc3dDUVlEVlFRRERBSk9RVEVkc0k1C0dDU3FHU01lM0RRRUUpB
U1lPZEdWemRFQmxiV0ZwYkM1amIyMHdIaGNOTWpVd056RXpNVGt5TURBNFdoY04KTWpZd056RXpNVGt5
TURBNFdqQjVNUXN3Q1FZRFZRUUdFd0pKVGpFU01CQUdBMVVFQ0F3S1NtaGhjbxRvWVc1awpNUkF3RGdZ
RFZRUUUEQWRfYUdGdVltRmtNUXN3Q1FZRFZRUUEtEQUppUVRFTE1Ba0dBMVVFQ3d3Q1RrRXhDekFKCk1Jn
TlZCQU1NQWw1Qk1SMHdHd1lKS29aSWh2Y05BUWtCRmc1MFpYTjBRR1Z0WVds0xtTnZiVENDQVNDd0RR
WUoKS29aSWh2Y05BUUVCQ1FBRGdnRVBIBRENDQVFvQ2dnRUJBS09kRk1ETnFsQW1MSUNjeTJDS2NBmjdE
YStkRVVqMgpWafR2NWVTUmZTNXZxKzJJSjNPaFJ5TXZHSEwzUmVxZEgwU1BpT0x6Y1BIOXB1MkFkVUJk
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K0UwK2RXdGtDNlQ2UFdJZFekUzRlTlRaelZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0Zl
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bmR5bytUNkZTSWErQUhFZ1VTV01seE8rUUpCCm1ZMG10VVOQkt2UWU3U3I1RSs4Yk5FMm5ROG45R0Fr
bURGUWhVb1RmQm1BMjVoRmthOFE3VjBDQXdfQUFhTlQKTUZFd0hRWURWUjBPQk1JRUZMS01BakZQbmJW
U2JSVzVWdU5PMUFAbnN5cGdNQjhHQTFVZE13UV1NQmFBRkxLTQpBakZQbmJWU2JSVzVWdU5PMUFAbnN5
cGdNQThHQTFVZE13UV1NQmFBRkxLTQpBakZQbmJWU2JSVzVWdU5PMUFAbnN5cGdNQThHQTFVZE13UV1NQm
MXpsU1FxdjBGaThLV3M5ZHRUK21NQTBud3o1WTY4Wjk1a2JlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0Zl
TjdJakxPSzN1VHI2QmpiTmRJMjZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0ZlZ0Zl
d2xpZitneUx4bndReFJkZ1hKdnoyYlFiMEk1ekxEU0NjdV1NNUtjcEw2UHU5K05EbGNveHJhZnVWCjNU
Mjg5UFVaZXhtSkUxenFrc3k1Z1l1seWl1amR4YUphTFBrUS9GVj12ew5aewM5UzdqQk1tSG10dER3OEJh
VlgKTUJWU20xQVd5ZzFpaUNxd1Bac0s3V3lwV0VYTy9sR1kvRmR3SVVIRn1DNVfQZTJ1c1B5QXFPMHZN
SnhDanF5MQo1S085bjNiKzM3L2dESkEvRXpKTH11TkxTeXB5a1BVPQotLS0tLUVORCBDRVJUSUZJQ0FUR
S0tLS0tCg==
```